Q1.Make a use of Toast and Explain with types.

Ans:

A Toast is a small popup message in Android used to show short feedback to the user without blocking the screen. It disappears automatically after a few seconds.

It is commonly used for showing success messages like “Data saved”, error messages like “Login failed”, quick information such as “No internet connection”, and also for debugging during development.

Syntax

Toast.makeText(getApplicationContext(), "Message", Toast.LENGTH_SHORT).show();

Types of Toast

1. Short Toast

Short Toast displays a message for a short duration of about 2 seconds.

Java

Toast.makeText(this, "Short Toast", Toast.LENGTH_SHORT).show();

2. Long Toast

Long Toast displays a message for a longer duration of about 3.5 to 4 seconds.

Java

Toast.makeText(this, "Long Toast", Toast.LENGTH_LONG).show();

3. Custom Toast

Custom Toast is used when you want to change layout, position, or appearance of the message.

Java

Toast toast = Toast.makeText(this, "Custom Toast", Toast.LENGTH_LONG);

toast.setGravity(Gravity.CENTER, 0, 0);

toast.show();

Or

Q2.Make a use of Dialogs and Explain with types

Ans:

A Dialog is a small popup window used to interact with the user. It blocks the screen until the user responds. It is used for confirmations, alerts, and taking input.

Types of Dialogs

1.Alert Dialog

Used to show message with buttons like OK or Cancel.

Java

AlertDialog.Builder builder = new AlertDialog.Builder(this);

Builder.setMessage("Are you sure?");

Builder.setPositiveButton("Yes", null);

Builder.show();

2.Progress Dialog

Used to show loading or processing.

Java

ProgressDialog progress = new ProgressDialog(this);

Progress.setMessage("Loading...");

Progress.show();

3.Custom Dialog

Used to create your own layout.

Java

Dialog dialog = new Dialog(this);

Dialog.setContentView(R.layout.custom_layout);

Dialog.show();

4.Date Picker Dialog

Used to select date.

Java

DatePickerDialog datePicker = new DatePickerDialog(this);

datePicker.show();

5.Time Picker Dialog

Used to select time.

Java

TimePickerDialog timePicker = new TimePickerDialog(this);

timePicker.show();

Q3.Make a use of Intent. Explain Passing Data using intent.

Ans:

An Intent is used to communicate between components in Android such as activities, services, or broadcast receivers. It is mainly used to start a new activity, open another app, or perform an action like calling or sending data.

Example:

Java

Intent intent = new Intent(this, SecondActivity.class);

startActivity(intent);

Passing Data using Intent

Data is passed using methods like putExtra() in the sending activity and received using getIntent() and getStringExtra() in the target activity.

Sending Data

Java

Intent intent = new Intent(this, SecondActivity.class);

intent.putExtra("name", "Junaid");

startActivity(intent);

Receiving Data

Java

```
Intent intent = getIntent();

String name = intent.getStringExtra("name");
```

Or

Q4.Make a use of Menus and explain with types

Ans:

Menus in Android are used to provide options or actions to the user in an application. They help organize features like settings, search, edit, and navigation in a structured way.

Types of Menus

1.Options Menu

This menu appears in the app bar or toolbar. It is used for main actions of the application.

Java

```
@Override

public boolean onCreateOptionsMenu(Menu menu) {

   getMenuInflater().inflate(R.menu.menu_main, menu);

   return true;

}
```

2.Context Menu

This menu appears when the user performs a long press on a view. It is used for item-specific actions.

Java

```
registerForContextMenu(textView);


@Override
```

```
public void onCreateContextMenu(ContextMenu menu, View v, ContextMenuInfo
menuInfo) {
   menu.add("Edit");
   menu.add("Delete");
}
```

3.Popup Menu

This menu appears near a specific view when triggered, usually by a button click.

Java

```
PopupMenu popup = new PopupMenu(this, view);
popup.getMenuInflater().inflate(R.menu.menu_main, popup.getMenu());
popup.show();
```

Q5.Define Data Storage. Explain Shared Preferences with example.

Ans:

Data Storage

Data Storage in Android means storing and managing app data so it can be used later. Data can be stored temporarily or permanently using different methods like Shared Preferences, files, or databases.

Shared Preferences

Shared Preferences is used to store small amounts of data in the form of key-value pairs. It is mainly used for saving simple data like user settings, login status, or preferences.

Example

Store Data

Java

```
SharedPreferences sp = getSharedPreferences(“MyData”, MODE_PRIVATE);
Sp.edit().putString(“name”, “Junaid”).apply();
```

Get Data

Java

String name = getSharedPreferences(“MyData”, MODE_PRIVATE).getString(“name”, “Default”);

Or

Q6.Define SQLite. Explain SQLiteOpenHelper Class.

Ans:

SQLite is a lightweight, built-in database in Android used to store structured data in tables. It does not require a separate server and is suitable for local data storage.

SQLiteOpenHelper Class

SQLiteOpenHelper is a helper class used to manage database creation and version control. It simplifies working with SQLite databases.

It provides methods to create and update the database.

Important Methods

onCreate() is used to create tables when database is first created.

onUpgrade() is used to update database when version changes.

getWritableDatabase() is used to insert or update data.

getReadableDatabase() is used to read data.

Q7.Explain Advantage and disadvantage of Sqlite

Ans:

Advantages

1. SQLite is lightweight and does not require a separate server.
2. It provides fast performance for small and medium data.
3. It supports standard SQL, so it is easy to use.
4. It stores the database in a single file, making it portable.
5. It requires very low setup and configuration.

Disadvantages

1. SQLite is not suitable for large-scale applications.
2. It allows only one write operation at a time (limited concurrency).
3. It does not provide advanced security features.
4. It lacks advanced database features.
5. Performance decreases with very large data.

Or

Q8. Explain Firebase. Explain Features of it

Ans:

Firebase is a platform developed by Google for building mobile and web applications. It provides backend services like database, authentication, and hosting, so developers do not need to manage servers.

Features

1. Realtime Database stores and syncs data instantly.
2. Authentication provides login using email, phone, or social accounts.
3. Cloud Firestore is a scalable database.
4. Cloud Storage stores files like images and videos.
5. Hosting is used to deploy apps.
6. Q9. Explain CRUD Operation

Q9.CRUD stands for Create, Read, Update, and Delete.

Ans:

It represents the basic operations performed on data in a database.

Create() is used to insert new data into the database.

Read() is used to retrieve or display data.

Update() is used to modify existing data.

Delete() is used to remove data from the database.

Or

Q10. Explain insert and retrieve perform in Firebase

Ans:

In Firebase, data is stored and accessed using a Realtime Database or Cloud Firestore.

Insert Data

Data is inserted using methods like setValue().

Java

```
DatabaseReference ref = FirebaseDatabase.getInstance().getReference("User");
ref.child("1").setValue("Junaid");
```

Retrieve Data

Data is retrieved using listeners like get() or addValueEventListener().

Java

```
DatabaseReference ref = FirebaseDatabase.getInstance().getReference("User");
ref.child("1").get().addOnSuccessListener(data -> {
   String name = data.getValue(String.class);
});
```

Q11. Update and Delete using SQLiteOpenHelper

Ans:

DBHelper

Java

```
Class DBHelper extends SQLiteOpenHelper {
   Public DBHelper(Context c) { super(c, “DB”, null, 1); }

   Public void onCreate(SQLiteDatabase db) {
      Db.execSQL(“CREATE TABLE user(id INTEGER, name TEXT)”);
   }
```

```
  Public void onUpgrade(SQLiteDatabase db, int o, int n) {}

  Void updateData(int id, String name) {
    ContentValues cv = new ContentValues();
    cv.put("name", name);
    getWritableDatabase().update("user", cv, "id=?", new String[]{id+""});
  }

  Void deleteData(int id) {
    getWritableDatabase().delete("user", "id=?", new String[]{id+""});
  }
}
```

Or

Q12. Create database and perform Insert & Read using SQLiteOpenHelper

Ans:

DBHelper

Java

```
Class DBHelper extends SQLiteOpenHelper {
  Public DBHelper(Context c) { super(c, "DB", null, 1); }

  Public void onCreate(SQLiteDatabase db) {
    Db.execSQL("CREATE TABLE user(id INTEGER, name TEXT)");
  }
```

```
    Public void onUpgrade(SQLiteDatabase db, int o, int n) {}


    Void insertData(int id, String name) {
        ContentValues cv = new ContentValues();
        cv.put("id", id);
        cv.put("name", name);
        getWritableDatabase().insert("user", null, cv);
    }


    Cursor getData() {
        Return getReadableDatabase().rawQuery("SELECT * FROM user", null);
    }
}
```

Q13. Define Kotlin. Write advantages of Kotlin

Ans:

Kotlin is a modern programming language used for Android development. It is officially supported by Google and runs on the Java Virtual Machine (JVM).

Advantages

1. Kotlin has concise syntax, so it reduces code length.
2. It is fully interoperable with Java, meaning both can be used together.
3. It provides null safety, reducing chances of crashes.
4. It supports modern features like data classes and lambdas.
5. It improves code readability and maintainability.

Q14. Explain Flutter with its features

Ans:

Flutter is an open-source UI toolkit developed by Google used to build cross-platform applications from a single codebase.

Features

1. It uses a single codebase for Android, iOS, and web apps.
2. It provides fast performance using native compilation.
3. It offers hot reload for quick UI changes during development.
4. It has rich UI widgets for attractive design.
5. It supports cross-platform development with high productivity.

Or

Q15. Explain React Native with its features

Ans:

React Native is a framework developed by Meta Platforms used to build mobile applications using JavaScript and React.

Features

1. It uses a single codebase for Android and iOS apps.
2. It provides near-native performance.
3. It supports reusable components.
4. It allows hot reload for fast development.
5. It uses JavaScript, making it easy for web developers.

Q16. Retrieve data and display in RecyclerView

Ans:

Get data from Firebase/JSON, store in list, and show using RecyclerView.

Java

```
List<String> list = new ArrayList<>();

DatabaseReference ref = FirebaseDatabase.getInstance().getReference(“User”);


Ref.addValueEventListener(new ValueEventListener() {
   Public void onDataChange(DataSnapshot s) {
      List.clear();
```

```
      For (DataSnapshot d : s.getChildren())
         List.add(d.getValue(String.class));


      recyclerView.setAdapter(new ArrayAdapter(this,
            android.R.layout.simple_list_item_1, list));
   }
   Public void onCancelled(DatabaseError e) {}
});
```